

Link Budget Justification & Operational Strategy for N4 Rocket Telemetry

Telemetry Systems Design Team

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1 Executive Summary

This document outlines the theoretical justification for achieving a telemetry link exceeding 5km range for a high-powered rocket flight. The system utilizes the Digi XBee-PRO 900HP (S3B) radio modules and standard 900MHz dipole “duck” antennas. Based on the link budget calculations presented below, the specified hardware possesses sufficient power margins to establish reliable communication at the target distance, provided that critical operational alignments are maintained to mitigate antenna radiation pattern limitations [1].

2 System Specifications

The core components of the telemetry link are defined by the following specifications.

Parameter	Value	Note
Frequency Band (f)	902–928 MHz	ISM Band (Center $\approx$ 915 MHz)
Transmitter Power (P_{tx})	+24 dBm	250 mW (XBee Pro S3B)
Receiver Sensitivity (R_x)	-110 dBm	At 10 Kbps Data Rate (Long Range Mode)
Tx Antenna Gain (G_{tx})	+2.0 dBi	SparkFun Duck Antenna
Rx Antenna Gain (G_{rx})	+2.0 dBi	SparkFun Duck Antenna
Target Distance (d)	5000 m	5 km Line of Sight

Table 1: System Link Parameters

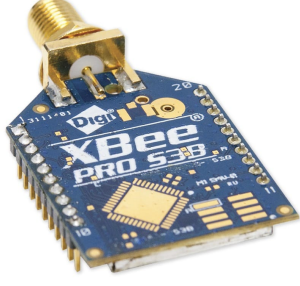


Figure 1: XBee-PRO 900HP S3B



Figure 2: 900MHz Duck Antenna

3 Theoretical Link Budget Calculation

To determine the viability of the link, we calculate the expected received signal strength (P_{rx}) using the Friis Transmission Equation [2].

3.1 Free Space Path Loss (FSPL)

The primary loss factor is the signal attenuation over distance in open air.

$$FSPL(dB) = 20 \log_{10}(d) + 20 \log_{10}(f) - 147.55 \quad (1)$$

Substituting our target parameters ($d = 5000\text{m}$, $f = 915 \times 10^6\text{Hz}$):

$$FSPL \approx 105.66 \text{ dB}$$

3.2 Link Margin Calculation

The expected received power (P_{rx}) is calculated by summing gains and subtracting losses:

$$P_{rx} = P_{tx} + G_{tx} + G_{rx} - FSPL \quad (2)$$

$$P_{rx} = 24 \text{ dBm} + 2 \text{ dBi} + 2 \text{ dBi} - 105.66 \text{ dB}$$

$$P_{rx} \approx -77.66 \text{ dBm}$$

The **Link Margin** is the safety buffer between the received signal and the receiver sensitivity (-110 dBm).

$$\text{Margin} = -77.66 \text{ dBm} - (-110 \text{ dBm}) \approx +\mathbf{32.34 \text{ dB}}$$

A margin of **+32.34 dB** justifies that the hardware is capable of distances significantly greater than 5km under ideal conditions [3].

4 Operational Challenges: The “Donut” Effect

Dipole antennas have significant nulls (blind spots) along their vertical axis. The radiation pattern resembles a donut, with the antenna sitting in the hole [4].

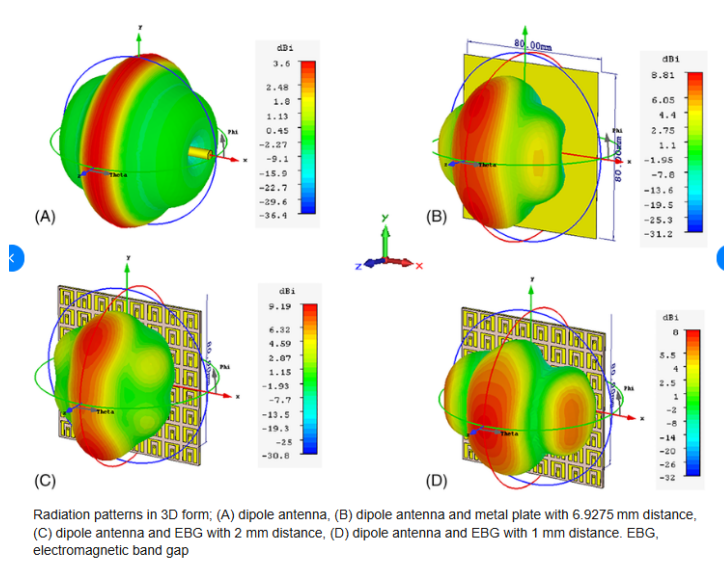


Figure 3: Typical 3D Radiation Pattern of a Dipole Antenna. Note the “nulls” along the Z-axis (Source: ResearchGate [4]).

The Risk: If the ground station antenna points vertically, and the rocket is overhead, the “nulls” align, causing signal loss.

5 Operational Strategy: Ground Station Alignment

To mitigate the null, the ground antenna should be tilted roughly 45° from vertical toward the launch pad.

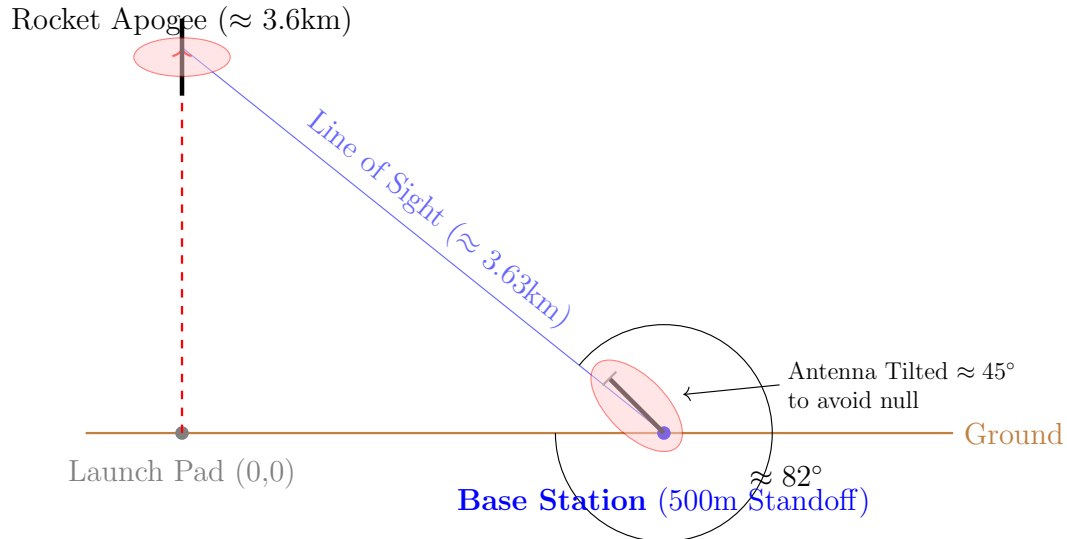


Figure 4: Ground Station Alignment Diagram. Tilting the base antenna ensures the rocket remains in the main radiation lobe during high-altitude flight.

A Post-Document Summary & Recommendations

A.1 Current Setup Issues

Despite the calculated 32dB margin, real-world factors can degrade performance:

- **The Nulls:** As detailed in Section 4, aligning the tips of the antennas causes signal dropout.
- **Polarization Mismatch:** The rocket antenna is vertically polarized. If the rocket tilts or spins during descent, polarization mismatch can cause losses of 3dB to 20dB [2].
- **Wasted Energy:** Omnidirectional antennas radiate energy into the ground and behind the station, wasting power that could be directed at the rocket.

A.2 Recommendations for Ground Station Upgrade

To ensure a robust 5km link, we recommend upgrading the **Ground Station antenna** while keeping the duck antenna on the rocket for aerodynamic efficiency.

Option 1: 900MHz Bi-Quad (Highly Recommended)

The Bi-Quad is often preferred by hobbyists over patch antennas because it is easier to build with standard tools and less sensitive to material errors.

- **Pros:** Excellent gain (≈ 11 dBi), wide beamwidth ($\approx 60^\circ$), forgiving fabrication tolerances.
- **Reference:** Changpuak Bi-Quad Calculator [6].

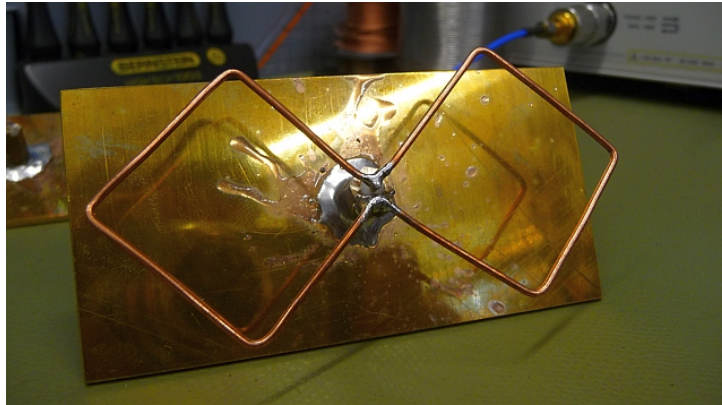


Figure 5: Example of a Homebrew 915MHz Bi-Quad Antenna.

Option 2: 900MHz Patch Antenna

The Patch Antenna is a strong candidate for this application due to its directional gain and flat form factor.

- **Pros:** Good gain (≈ 7 dBi), robust flat form factor, easier to mount on a tripod than a Yagi.

- **Cons:** Extremely sensitive to material variations (FR-4 fiberglass quality).
- **Design Resource:** See Phil’s Lab for detailed PCB patch design procedures [5].

A.3 Crucial Tooling

For any DIY antenna build (Patch or Bi-Quad), a **NanoVNA** is mandatory. Without verifying the resonant frequency, the antenna may be tuned to the wrong frequency (e.g., 880MHz), significantly reducing range.

References

- [1] Digi International, “XBee-PRO 900HP RF Modules User Guide,” 2016. [Online]. Available: <https://hub.digi.com/dp/path=/support/asset/xbee-pro-900hp-user-guide/>
- [2] D. M. Pozar, *Microwave Engineering*, 4th ed., Wiley, 2011, pp. 600-605.
- [3] H. Ward Silver, Ed., *The ARRL Antenna Book for Radio Communications*, 24th ed., ARRL, 2019, ch. 4.
- [4] P. Muthu Kannan, et al., “3D Radiation pattern of normal dipole antenna,” *ResearchGate*, 2020. [Online]. Available: https://www.researchgate.net/figure/D-Radiation-pattern-of-normal-dipole-antenna_fig2_347276602
- [5] Phil’s Lab, “PCB Antenna Design, Simulation, and Fabrication,” YouTube, 2021. [Online]. Available: <https://www.youtube.com/watch?v=UQBMR0v7Dy4>
- [6] Changpuak.ch, “Bi-Quad Antenna Calculator,” [Online]. Available: https://www.changpuak.ch/electronics/bi_quad_antenna_designer.php